

Unit III – Random Processes

(Notes by Dr. S. Balaji)

1. Strict-Sense Stationary (SSS) Process

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A random process $\{X(t)\}$ is **SSS** if its statistical properties remain unchanged by any shift c of the time origin, i.e. for all n and c ,

$$f(x_1, \dots, x_n; t_1, \dots, t_n) = f(x_1, \dots, x_n; t_1 + c, \dots, t_n + c).$$

Consequently,

$$(i) \quad E[X(t)] = \text{constant}, \quad (ii) \quad \text{Var}[X(t)] = \text{constant}.$$

In the problems below, (i) and (ii) are used as the working test for stationarity.

Wide-Sense Stationary (WSS) Process

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A random process is **WSS** if only its first and second-order moments are time-invariant:

$$(i) \quad E[X(t)] = \text{constant},$$

$$(ii) \quad R_X(t_1, t_2) = R_X(t_1 - t_2) = R_X(\tau), \quad \tau = t_1 - t_2,$$

i.e. R_X depends only on the time difference τ .

A random process $\{X(t)\}$ which is not stationary is called an **evolutionary process**.

2. Definition: Uniform Distribution

Uniform Distribution

A random variable X is said to follow a **uniform distribution** in (a, b) if its pdf is

$$f(x) = \begin{cases} \frac{1}{b-a}, & a < x < b, \\ 0, & \text{otherwise.} \end{cases}$$

Then X is uniformly distributed on (a, b) .

3. Problem 1 – Non-Stationary Random Process

Question

Customers arrive at a counter during time t minutes. Let $\{X(t)\}$ be a random process with probability distribution

$$P(X(t) = n) = \begin{cases} \frac{at}{1+at}, & n = 0, \\ \frac{(at)^{n-1}}{(1+at)^{n+1}}, & n = 1, 2, 3, \dots \end{cases}$$

where a is a constant. **Show that $\{X(t)\}$ is not stationary.**

Solution

The $n = 0$ term contributes $0 \cdot \frac{at}{1+at} = 0$, so

$$\begin{aligned} E[X(t)] &= \sum_{n=0}^{\infty} n P(X(t) = n) = \sum_{n=1}^{\infty} n \frac{(at)^{n-1}}{(1+at)^{n+1}} \\ &= \frac{1}{(1+at)^2} \sum_{n=1}^{\infty} n \left(\frac{at}{1+at} \right)^{n-1} \\ &= \frac{1}{(1+at)^2} \left[1 + 2 \left(\frac{at}{1+at} \right) + 3 \left(\frac{at}{1+at} \right)^2 + \dots \right]. \end{aligned}$$

Let $y = \frac{at}{1+at}$ ($0 < y < 1$). Using $1 + 2y + 3y^2 + \dots = (1-y)^{-2}$,

$$E[X(t)] = \frac{1}{(1+at)^2} (1-y)^{-2}.$$

Substituting y :

$$1 - y = 1 - \frac{at}{1+at} = \frac{1+at-at}{1+at} = \frac{1}{1+at} \Rightarrow (1-y)^{-2} = (1+at)^2.$$

$$\therefore E[X(t)] = \frac{1}{(1+at)^2} (1+at)^2 = 1 \Rightarrow \boxed{E[X(t)] = 1, \text{ a constant.}}$$

(ii) Variance of $X(t)$

Since $E[X(t)] = 1$,

$$\begin{aligned} \text{Var}[X(t)] &= E[X^2(t)] - (E[X(t)])^2 = E[X^2(t)] - 1 \\ &= \sum_n n^2 P(X(t) = n) - 1 \\ &= \sum_n (n^2 + n - n) P(X(t) = n) - 1. \end{aligned}$$

$$\begin{aligned} \text{Var}[X(t)] &= \sum_n n(n+1) P(X(t) = n) - \sum_n n P(X(t) = n) - 1 \\ &= \sum_n n(n+1) P(X(t) = n) - 1 - 1 \\ &= \sum_n n(n+1) P(X(t) = n) - 2, \end{aligned}$$

since $\sum_n n P(X(t) = n) = E[X(t)] = 1$.

Substituting the given probabilities (the $n = 0$ term is zero):

$$\begin{aligned} \text{Var}[X(t)] &= \sum_{n=1}^{\infty} n(n+1) \frac{(at)^{n-1}}{(1+at)^{n+1}} - 2 \\ &= \frac{1}{(1+at)^2} [1 \cdot 2 + 2 \cdot 3y + 3 \cdot 4y^2 + \dots] - 2 \\ &= \frac{2}{(1+at)^2} [1 + 3y + 6y^2 + \dots] - 2, \quad y = \frac{at}{1+at}. \end{aligned}$$

Using $1 + 3y + 6y^2 + \dots = (1-y)^{-3}$:

$$\begin{aligned} \text{Var}[X(t)] &= \frac{2}{(1+at)^2} (1-y)^{-3} - 2 = \frac{2}{(1+at)^2} \left(\frac{1}{1+at} \right)^{-3} - 2 \\ &= \frac{2}{(1+at)^2} (1+at)^3 - 2 = 2(1+at) - 2 = 2at. \end{aligned}$$

$$\boxed{\text{Var}[X(t)] = 2at}$$

$E[X(t)]$ is constant but $\text{Var}[X(t)]$ depends on t .

$\therefore \{X(t)\}$ is a Non-Stationary (Evolutionary) Process.

Problem 2 – Stationarity of $X(t) = \cos(\omega_0 t + \theta)$

Question

A radio station broadcasts its signal in the form of a cosine wave. Receivers at different places receive signals modelled as

$$X(t) = \cos(\omega_0 t + \theta),$$

where ω_0 is a constant and θ is a uniformly distributed random variable in $(-\pi, \pi)$.

Examine the stationarity of $\{X(t)\}$.

Solution

The pdf of θ is $f(\theta) = \frac{1}{2\pi}$, $-\pi < \theta < \pi$ (and 0 otherwise).

$$\begin{aligned} E[X(t)] &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \cos(\omega_0 t + \theta) d\theta = \frac{1}{2\pi} \left[\sin(\omega_0 t + \theta) \right]_{-\pi}^{\pi} \\ &= \frac{1}{2\pi} \left[\sin(\omega_0 t + \pi) - \sin(\omega_0 t - \pi) \right] \\ &= \frac{1}{2\pi} \left[-\sin \omega_0 t - (-\sin \omega_0 t) \right] = 0. \end{aligned}$$

$$\boxed{E[X(t)] = 0, \text{ a constant.}}$$

(ii) Variance of $X(t)$

$$\begin{aligned}
\text{Var}[X(t)] &= E[X^2(t)] - (E[X(t)])^2 = E[X^2(t)] \\
&= \frac{1}{2\pi} \int_{-\pi}^{\pi} \cos^2(\omega_0 t + \theta) d\theta \\
&= \frac{1}{4\pi} \int_{-\pi}^{\pi} [1 + \cos(2\omega_0 t + 2\theta)] d\theta \\
&= \frac{1}{4\pi} \left[\theta + \frac{\sin(2\omega_0 t + 2\theta)}{2} \right]_{-\pi}^{\pi}
\end{aligned}$$

$$\begin{aligned}
\text{Var}[X(t)] &= \frac{1}{4\pi} \left[\pi - (-\pi) + \frac{1}{2} (\sin(2\omega_0 t + 2\pi) - \sin(2\omega_0 t - 2\pi)) \right] \\
&= \frac{1}{4\pi} \left[2\pi + \frac{1}{2} (\sin 2\omega_0 t - \sin 2\omega_0 t) \right] \\
&= \frac{1}{4\pi} [2\pi + 0] = \frac{1}{2}.
\end{aligned}$$

$$\text{Var}[X(t)] = \frac{1}{2}, \text{ a constant.}$$

Conclusion

Thus both the mean and the variance are constants:

$$X(t) = \cos(\omega_0 t + \theta) \text{ is a Stationary Random Process.}$$

Problem 3 – Stationarity of $X(t) = A \cos(\omega_0 t + \theta)$ **Question**

Show that the random process

$$X(t) = A \cos(\omega_0 t + \theta)$$

is a **Stationary Random Process**, where A and ω_0 are constants and θ is a uniformly distributed random variable in the interval $(0, 2\pi)$.

Solution

The pdf of θ is $f(\theta) = \frac{1}{2\pi}$, $0 < \theta < 2\pi$ (and 0 otherwise).

$$\begin{aligned}
E[X(t)] &= \int_0^{2\pi} A \cos(\omega_0 t + \theta) \frac{1}{2\pi} d\theta = \frac{A}{2\pi} \left[\sin(\omega_0 t + \theta) \right]_0^{2\pi} \\
&= \frac{A}{2\pi} \left[\sin(\omega_0 t + 2\pi) - \sin(\omega_0 t + 0) \right] \\
&= \frac{A}{2\pi} \left[\sin \omega_0 t - \sin \omega_0 t \right] = 0.
\end{aligned}$$

$$E[X(t)] = 0, \text{ a constant (not a function of } t \text{).}$$

(ii) Variance of $X(t)$

$$\begin{aligned}
\text{Var}[X(t)] &= E[X^2(t)] - (E[X(t)])^2 = E[X^2(t)] \\
&= \frac{A^2}{2\pi} \int_0^{2\pi} \cos^2(\omega_0 t + \theta) d\theta \\
&= \frac{A^2}{2\pi} \int_0^{2\pi} \frac{1 + \cos(2\omega_0 t + 2\theta)}{2} d\theta \\
&= \frac{A^2}{4\pi} \left[\theta + \frac{\sin(2\omega_0 t + 2\theta)}{2} \right]_0^{2\pi}
\end{aligned}$$

$$\begin{aligned}
\text{Var}[X(t)] &= \frac{A^2}{4\pi} \left[(2\pi - 0) + \frac{1}{2} (\sin(2\omega_0 t + 4\pi) - \sin(2\omega_0 t + 0)) \right] \\
&= \frac{A^2}{4\pi} \left[2\pi + \frac{1}{2} (\sin 2\omega_0 t - \sin 2\omega_0 t) \right] \\
&= \frac{A^2}{4\pi} [2\pi + 0] = \frac{A^2}{2}.
\end{aligned}$$

$$\boxed{\text{Var}[X(t)] = \frac{A^2}{2}, \text{ a constant.}}$$

Conclusion

Since both $E[X(t)]$ and $\text{Var}[X(t)]$ are constants (independent of t),

$$\boxed{\therefore X(t) = A \cos(\omega_0 t + \theta) \text{ is a Stationary Random Process.}}$$

Example – Stationarity of $X(t) = A \sin(\omega t + \phi)$ **Question**

Check the stationarity of the collection of signals given by

$$X(t) = A \sin(\omega t + \phi),$$

where A and ω are constants, and ϕ is a uniformly distributed random variable in $(-\pi, \pi)$.

Solution

The pdf of ϕ is $f(\phi) = \frac{1}{2\pi}$, $-\pi < \phi < \pi$ (and 0 otherwise).

$$\begin{aligned}
E[X(t)] &= \frac{A}{2\pi} \int_{-\pi}^{\pi} \sin(\omega t + \phi) d\phi = \frac{A}{2\pi} \left[-\cos(\omega t + \phi) \right]_{-\pi}^{\pi} \\
&= \frac{A}{2\pi} \left[-\cos(\omega t + \pi) + \cos(\omega t - \pi) \right] \\
&= \frac{A}{2\pi} \left[\cos \omega t - \cos \omega t \right] = 0.
\end{aligned}$$

$$\boxed{E[X(t)] = 0, \text{ a constant.}}$$

(ii) Variance of $X(t)$

Since $E[X(t)] = 0$,

$$\begin{aligned}\text{Var}[X(t)] &= E[X^2(t)] = \frac{A^2}{2\pi} \int_{-\pi}^{\pi} \sin^2(\omega t + \phi) d\phi \\ &= \frac{A^2}{2\pi} \int_{-\pi}^{\pi} \frac{1 - \cos(2\omega t + 2\phi)}{2} d\phi \\ &= \frac{A^2}{4\pi} \left[\phi - \frac{\sin(2\omega t + 2\phi)}{2} \right]_{-\pi}^{\pi}\end{aligned}$$

$$\begin{aligned}\text{Var}[X(t)] &= \frac{A^2}{4\pi} \left[(\pi - (-\pi)) - \frac{1}{2} (\sin(2\omega t + 2\pi) - \sin(2\omega t - 2\pi)) \right] \\ &= \frac{A^2}{4\pi} \left[2\pi - \frac{1}{2} (\sin 2\omega t - \sin 2\omega t) \right] \\ &= \frac{A^2}{4\pi} [2\pi - 0] = \frac{A^2}{2}.\end{aligned}$$

$$\boxed{\text{Var}[X(t)] = \frac{A^2}{2}, \text{ a constant.}}$$

Conclusion

Since both $E[X(t)]$ and $\text{Var}[X(t)]$ are constants (independent of t),

$$\boxed{\therefore X(t) = A \sin(\omega t + \phi) \text{ is a Stationary Random Process.}}$$

Recall: WSS Process

$$(i) E[X(t)] = \text{constant}, \quad (ii) R_X(t_1, t_2) = R_X(t_1 - t_2) = R_X(\tau).$$

Problem – $X(t) = A \cos(\omega_0 t + \theta)$ is a WSS Process**Question**

Prove that the random process

$$X(t) = A \cos(\omega_0 t + \theta)$$

is a **Wide-Sense Stationary (WSS)** process, where A and ω_0 are constants and θ is a uniformly distributed random variable in $(0, 2\pi)$.

Solution

The pdf of θ is $f(\theta) = \frac{1}{2\pi}$, $0 < \theta < 2\pi$ (and 0 otherwise).

$$\begin{aligned} E[X(t)] &= \int_0^{2\pi} A \cos(\omega_0 t + \theta) \frac{1}{2\pi} d\theta = \frac{A}{2\pi} \left[\sin(\omega_0 t + \theta) \right]_0^{2\pi} \\ &= \frac{A}{2\pi} \left[\sin(\omega_0 t + 2\pi) - \sin(\omega_0 t + 0) \right] \\ &= \frac{A}{2\pi} \left[\sin \omega_0 t - \sin \omega_0 t \right] = 0. \end{aligned}$$

$$\boxed{E[X(t)] = 0, \text{ a constant.}}$$

(ii) ACF of $X(t)$

Using $\cos A \cos B = \frac{1}{2} [\cos(A + B) + \cos(A - B)]$:

$$\begin{aligned} R_X(t_1, t_2) &= E[X(t_1)X(t_2)] \\ &= A^2 E[\cos(\omega_0 t_1 + \theta) \cos(\omega_0 t_2 + \theta)] \\ &= \frac{A^2}{2} E[\cos(\omega_0(t_1 + t_2) + 2\theta) + \cos(\omega_0(t_1 - t_2))] \\ &= \frac{A^2}{2} \left[\frac{1}{2\pi} \int_0^{2\pi} \cos(\omega_0(t_1 + t_2) + 2\theta) d\theta \right. \\ &\quad \left. + \cos(\omega_0(t_1 - t_2)) \right]. \end{aligned}$$

$$\begin{aligned} \int_0^{2\pi} \cos(\omega_0(t_1 + t_2) + 2\theta) d\theta &= \frac{1}{2} \left[\sin(\omega_0(t_1 + t_2) + 2\theta) \right]_0^{2\pi} \\ &= \frac{1}{2} \left[\sin(\omega_0(t_1 + t_2) + 4\pi) - \sin(\omega_0(t_1 + t_2) + 0) \right] \\ &= \frac{1}{2} \left[\sin \omega_0(t_1 + t_2) - \sin \omega_0(t_1 + t_2) \right] = 0. \end{aligned}$$

$$\therefore R_X(t_1, t_2) = \frac{A^2}{2} [0 + \cos(\omega_0(t_1 - t_2))] = \frac{A^2}{2} \cos[\omega_0(t_1 - t_2)].$$

Conclusion

With $\tau = t_1 - t_2$,

$$\boxed{R_X(t_1, t_2) = \frac{A^2}{2} \cos[\omega_0(t_1 - t_2)] = \frac{A^2}{2} \cos \omega_0 \tau = R_X(\tau).}$$

Hence $R_X(t_1, t_2)$ is a function of the time difference only.

$$\boxed{\begin{aligned} E[X(t)] &= \text{constant}, \\ R_X(t_1, t_2) &= R_X(t_1 - t_2). \end{aligned}}$$

$\therefore X(t) = A \cos(\omega_0 t + \theta)$ is a **WSS Process**.

Problem – $Y(t) = X(t) \cos(\omega t + \phi)$ is a WSS Process

Question

Given the random process

$$Y(t) = X(t) \cos(\omega t + \phi),$$

where $X(t)$ is a WSS process, ω is a constant, and ϕ is a uniformly distributed random variable in $(-\pi, \pi)$, independent of $X(t)$, prove that $Y(t)$ is also a WSS process.

Solution

Since $X(t)$ is WSS:

- (i) $E[X(t)] = \text{constant} = k$,
- (ii) $R_X(t_1, t_2) = R_X(t_1 - t_2) = h(t_1 - t_2)$.

Also, ϕ is uniformly distributed in $(-\pi, \pi)$ and is independent of $X(t)$:

$$f(\phi) = \begin{cases} \frac{1}{2\pi}, & -\pi < \phi < \pi, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Mean of $Y(t)$

Since $X(t)$ and ϕ are independent,

$$\begin{aligned} E[Y(t)] &= E[X(t) \cos(\omega t + \phi)] = E[X(t)] E[\cos(\omega t + \phi)] \\ &= \frac{k}{2\pi} \left[\sin(\omega t + \phi) \right]_{-\pi}^{\pi} \\ &= \frac{k}{2\pi} \left[\sin(\omega t + \pi) - \sin(\omega t - \pi) \right] \\ &= \frac{k}{2\pi} \left[-\sin \omega t + \sin \omega t \right] = 0. \end{aligned}$$

$$\boxed{E[Y(t)] = 0, \text{ a constant.}}$$

(b) ACF of $Y(t)$

Since $X(t)$ and ϕ are independent,

$$\begin{aligned} R_Y(t_1, t_2) &= E[Y(t_1)Y(t_2)] \\ &= E[X(t_1) \cos(\omega t_1 + \phi) X(t_2) \cos(\omega t_2 + \phi)] \\ &= E[X(t_1)X(t_2)] E[\cos(\omega t_1 + \phi) \cos(\omega t_2 + \phi)] \\ &= R_X(t_1, t_2) E[\cos(\omega t_1 + \phi) \cos(\omega t_2 + \phi)]. \end{aligned}$$

Using $\cos A \cos B = \frac{1}{2}[\cos(A + B) + \cos(A - B)]$:

$$\begin{aligned} E[\cos(\omega t_1 + \phi) \cos(\omega t_2 + \phi)] &= \frac{1}{2} E[\cos(\omega(t_1 + t_2) + 2\phi) + \cos(\omega(t_1 - t_2))] \\ &= \frac{1}{2} \left[\frac{1}{2\pi} \int_{-\pi}^{\pi} \cos(\omega(t_1 + t_2) + 2\phi) d\phi \right. \\ &\quad \left. + \cos(\omega(t_1 - t_2)) \right]. \end{aligned}$$

$$\begin{aligned}
\int_{-\pi}^{\pi} \cos(\omega(t_1 + t_2) + 2\phi) d\phi &= \frac{1}{2} \left[\sin(\omega(t_1 + t_2) + 2\phi) \right]_{-\pi}^{\pi} \\
&= \frac{1}{2} \left[\sin(\omega(t_1 + t_2) + 2\pi) - \sin(\omega(t_1 + t_2) - 2\pi) \right] \\
&= \frac{1}{2} \left[\sin \omega(t_1 + t_2) - \sin \omega(t_1 + t_2) \right] = 0.
\end{aligned}$$

$$\therefore E[\cos(\omega t_1 + \phi) \cos(\omega t_2 + \phi)] = \frac{1}{2} \cos[\omega(t_1 - t_2)].$$

Conclusion

Substituting back,

$$R_Y(t_1, t_2) = R_X(t_1, t_2) \cdot \frac{1}{2} \cos[\omega(t_1 - t_2)] = \frac{1}{2} R_X(t_1 - t_2) \cos[\omega(t_1 - t_2)].$$

$$R_Y(t_1, t_2) = \frac{1}{2} R_X(\tau) \cos \omega \tau = R_Y(\tau), \quad \tau = t_1 - t_2.$$

$$(i) \quad E[Y(t)] = \text{constant},$$

$$(ii) \quad R_Y(t_1, t_2) = R_Y(t_1 - t_2).$$

$\therefore Y(t)$ is a **WSS Process**.

Exercise**Exercise Problem**

Question: Given the random process $X(t) = A \cos(\omega_0 t + \theta)$, where A and ω_0 are constants and θ is uniformly distributed in $(0, \pi)$, **check the stationarity of the random process.**

Answer: Here $f(\theta) = \frac{1}{\pi}$, $0 < \theta < \pi$, and

$$\begin{aligned}
E[X(t)] &= \frac{A}{\pi} \left[\sin(\omega_0 t + \theta) \right]_0^{\pi} = \frac{A}{\pi} \left[\sin(\omega_0 t + \pi) - \sin(\omega_0 t + 0) \right] \\
&= \frac{A}{\pi} \left[-\sin \omega_0 t - \sin \omega_0 t \right] = -\frac{2A}{\pi} \sin \omega_0 t,
\end{aligned}$$

which depends on t . Hence $X(t)$ is **not stationary**.

Problem – $X(t) = A \cos \omega t + B \sin \omega t$ is a WSS Process**Question**

Show that the random process

$$X(t) = A \cos \omega t + B \sin \omega t$$

is a WSS process, where A and B are independent random variables with mean zero and variance 1.

Solution

A and B are independent with mean 0 and variance 1:

$$E(A) = 0, \quad E(B) = 0. \quad (1)$$

$$\text{Var}(A) = 1 = E(A^2) - (E(A))^2,$$

$$\text{Var}(B) = 1 = E(B^2) - (E(B))^2.$$

$$\text{Using (1): } E(A^2) = 1, \quad E(B^2) = 1. \quad (2)$$

$$\text{Since } A \text{ and } B \text{ are independent: } E(AB) = E(A)E(B) = 0 = E(BA). \quad (3)$$

Mean and ACF

Mean: $E[X(t)] = E(A) \cos \omega t + E(B) \sin \omega t = 0$ (constant).

ACF:

$$\begin{aligned} R_X(t_1, t_2) &= E[(A \cos \omega t_1 + B \sin \omega t_1)(A \cos \omega t_2 + B \sin \omega t_2)] \\ &= E(A^2) \cos \omega t_1 \cos \omega t_2 + E(AB) \cos \omega t_1 \sin \omega t_2 \\ &\quad + E(BA) \sin \omega t_1 \cos \omega t_2 + E(B^2) \sin \omega t_1 \sin \omega t_2 \\ &= \cos \omega t_1 \cos \omega t_2 + \sin \omega t_1 \sin \omega t_2 \quad (\text{by (2), (3)}) \\ &= \cos[\omega(t_1 - t_2)]. \end{aligned}$$

Conclusion

$$E[X(t)] = 0 \text{ (constant)}$$

$$R_X(t_1, t_2) = \cos[\omega(t_1 - t_2)] = \cos \omega \tau = R_X(\tau)$$

$$\begin{aligned} (i) \quad &E[X(t)] = \text{constant}, \\ (ii) \quad &R_X(t_1, t_2) = \text{a function of time difference.} \end{aligned}$$

$$\text{Therefore, } X(t) = A \cos \omega t + B \sin \omega t \text{ is WSS.}$$

Problem – $X(t) = Y \cos \omega t + Z \sin \omega t$ is a WSS Process**Question**

Show that the random process

$$X(t) = Y \cos \omega t + Z \sin \omega t$$

is a WSS process, where Y and Z are independent random variables which assume the values -2 and 1 with probabilities $\frac{1}{3}$ and $\frac{2}{3}$ respectively.

Solution

$$\begin{array}{c|c|c} y & -2 & 1 \\ \hline P(Y=y) & \frac{1}{3} & \frac{2}{3} \end{array} \qquad \begin{array}{c|c|c} z & -2 & 1 \\ \hline P(Z=z) & \frac{1}{3} & \frac{2}{3} \end{array}$$

$$E(Y) = (-2) \left(\frac{1}{3}\right) + (1) \left(\frac{2}{3}\right) = 0; \text{ similarly } E(Z) = 0. \quad (1)$$

$$E(Y^2) = (-2)^2 \left(\frac{1}{3}\right) + 1^2 \left(\frac{2}{3}\right) = \frac{4}{3} + \frac{2}{3} = 2; \text{ sim. } E(Z^2) = 2. \quad (2)$$

$$E(YZ) = E(Y)E(Z) = 0 = E(ZY) \quad (\text{independence}). \quad (3)$$

Mean and ACF

Mean: $E[X(t)] = E(Y) \cos \omega t + E(Z) \sin \omega t = 0$ (by (1)).

ACF:

$$\begin{aligned} R_X(t_1, t_2) &= E[(Y \cos \omega t_1 + Z \sin \omega t_1)(Y \cos \omega t_2 + Z \sin \omega t_2)] \\ &= E(Y^2) \cos \omega t_1 \cos \omega t_2 + E(YZ) \cos \omega t_1 \sin \omega t_2 \\ &\quad + E(ZY) \sin \omega t_1 \cos \omega t_2 + E(Z^2) \sin \omega t_1 \sin \omega t_2 \\ &= 2 \cos \omega t_1 \cos \omega t_2 + 0 + 0 + 2 \sin \omega t_1 \sin \omega t_2 \\ &= 2 \cos[\omega(t_1 - t_2)] \quad (\text{by (2), (3)}). \end{aligned}$$

Conclusion

$$E[X(t)] = 0, \text{ constant}$$

$$R_X(t_1, t_2) = 2 \cos[\omega(t_1 - t_2)] = R_X(\tau)$$

Since

1. $E[X(t)]$ is a constant, and
2. $R_X(t_1, t_2)$ depends only on the time difference $t_1 - t_2$,

$$\therefore X(t) \text{ is a WSS Process.}$$

Exercise

Exercise Problem

Consider the random process

$$X(t) = A \cos \omega t + B \sin \omega t,$$

where A and B are uncorrelated random variables with zero mean and the same variance σ^2 . **Prove that $X(t)$ is a WSS process.**

Hint: $E(AB) = E(A)E(B) = 0$ (uncorrelated), $E(A^2) = E(B^2) = \sigma^2$; show that $E[X(t)] = 0$ and $R_X(\tau) = \sigma^2 \cos \omega \tau$.

Definitions

Cross-Correlation Function (CCF)

The **CCF** of two random processes $\{X(t)\}$ and $\{Y(t)\}$ is $R_{XY}(t_1, t_2) = E[X(t_1)Y(t_2)]$. If the processes are jointly stationary, it is a function of the time difference only.

Jointly WSS Processes

$\{X(t)\}$ and $\{Y(t)\}$ are **jointly WSS** if (1) each is individually WSS, and (2) $R_{XY}(t_1, t_2)$ is a function of the time difference only.

Autocorrelation Function (ACF) – Definition**Definition**

Let $\{x(t)\}$ be a wide-sense stationary random process. Its autocorrelation function (ACF) depends only on the time difference $\tau = t_1 - t_2$:

$$\begin{aligned} R(\tau) &= E[x(t_1)x(t_2)], \quad \tau = t_1 - t_2, \\ &= E[x(t)x(t - \tau)]. \end{aligned}$$

Thus $R(\tau)$ indicates how similar two signal values are when separated by τ units of time.

Properties of the ACF – Property 1**Property 1**

$R(\tau)$ is an even function of τ : $R(\tau) = R(-\tau)$.

Proof

By definition, $R(\tau) = E[x(t)x(t - \tau)]$, so $R(-\tau) = E[x(t)x(t + \tau)]$.
Let $\lambda = t + \tau \Rightarrow t = \lambda - \tau$:

$$\begin{aligned} R(-\tau) &= E[x(\lambda - \tau)x(\lambda)] = E[x(\lambda)x(\lambda - \tau)] \\ &= R(\tau) \quad (\text{WSS: independent of } \lambda). \end{aligned}$$

Hence $R(\tau)$ is even.

Properties of the ACF – Property 2**Property 2**

The ACF is bounded by its zero-lag value: $|R(\tau)| \leq R(0)$.

Proof

$R(\tau) = E[x(t_1)x(t_2)]$ with $\tau = t_1 - t_2$, and $R(0) = E[x^2(t)]$ for every t .
Consider $E[(x(t_1) + x(t_2))^2] \geq 0$. Expanding and using stationarity,

$$\begin{aligned} E[x^2(t_1)] + E[x^2(t_2)] + 2E[x(t_1)x(t_2)] &\geq 0 \\ 2R(0) + 2R(\tau) &\geq 0 \Rightarrow R(\tau) \geq -R(0). \end{aligned} \tag{1}$$

Similarly, from $E[(x(t_1) - x(t_2))^2] \geq 0$,

$$\begin{aligned} E[x^2(t_1)] + E[x^2(t_2)] - 2E[x(t_1)x(t_2)] &\geq 0 \\ 2R(0) - 2R(\tau) &\geq 0 \Rightarrow R(\tau) \leq R(0). \end{aligned} \tag{2}$$

Combining (1) and (2):

$$-R(0) \leq R(\tau) \leq R(0) \implies |R(\tau)| \leq R(0).$$

Properties of the ACF – Property 3

Property 3

If $x(t)$ has mean μ_x (and no periodic component), then $\lim_{\tau \rightarrow \infty} R(\tau) = \mu_x^2$.

Proof

$R(\tau) = E[x(t)x(t-\tau)]$. As $\tau \rightarrow \infty$, the values $x(t)$ and $x(t-\tau)$ become statistically independent, so

$$\lim_{\tau \rightarrow \infty} R(\tau) = E[x(t)] \cdot E[x(t-\tau)] = \mu_x \cdot \mu_x = \mu_x^2.$$

Problem 1 – Suitable ACF

Problem 1

Which one of the following is a suitable ACF?

- (i) $A \sin(\omega\tau)$ (ii) $A \cos(\omega\tau)$

Solution

An ACF $R(\tau)$ must be an even function of τ . Cosine is even, while sine is odd, so $A \sin(\omega\tau)$ is not an ACF.

For $R(\tau) = A \cos(\omega\tau)$ with $A > 0$: $R(0) = A \cos 0 = A$ and

$$|R(\tau)| = A |\cos(\omega\tau)| \leq A = R(0),$$

which satisfies the ACF property. $\therefore A \cos(\omega\tau)$ is a valid ACF.

Problem 2 – Mean and Variance from ACF

Problem 2

Given the ACF of a stationary R.P. $\{x(t)\}$ as $R(\tau) = 25 + \frac{4}{1+6\tau^2}$, find the mean and variance of the process.

Solution

Using $\lim_{\tau \rightarrow \infty} R(\tau) = \mu_x^2$:

$$\begin{aligned} \mu_x^2 &= \lim_{\tau \rightarrow \infty} \left(25 + \frac{4}{1+6\tau^2} \right) = 25 + \frac{4}{1+6(\infty)^2} \\ &= 25 + 0 = 25 \implies |\mu_x| = 5. \end{aligned}$$

So $\mu_x = \pm 5$ (the ACF fixes only μ_x^2 ; we take $\mu_x = 5$).

Variance: $\text{Var}(x(t)) = E[x^2(t)] - \mu_x^2$, where

$$E[x^2(t)] = R(0) = 25 + \frac{4}{1 + 6(0)^2} = 25 + 4 = 29.$$

$$\text{Var}(x(t)) = 29 - 25 = 4.$$

$$\therefore \boxed{\mu_x = 5 \text{ (or } -5), \quad \sigma_x^2 = 4.}$$

Problem 3 – Mean, Variance and Covariance

Problem 3

Let $x(t)$ be a stationary R.P. with mean 3 and ACF $R_x(t_1, t_2) = 9 + 4e^{-0.2|t_1 - t_2|}$. Determine the mean, variance, and covariance of the random variables $X(5)$ and $X(8)$.

Solution

Since the process is stationary,

$$R(\tau) = 9 + 4e^{-0.2|\tau|}, \quad \tau = t_1 - t_2, \quad E[x(t)] = \mu_x = 3 \quad \forall t.$$

Let $Z = X(5)$ and $W = X(8)$. Then $E[Z] = E[X(5)] = 3$ and $E[W] = E[X(8)] = 3$.

Variance:

$$E[Z^2] = R(0) = 9 + 4e^{-0.2(0)} = 9 + 4 = 13,$$

$$\text{Var}(Z) = E[Z^2] - (E[Z])^2 = 13 - 3^2 = 13 - 9 = 4.$$

Similarly, $\text{Var}(W) = 4$.

Covariance: Since $|5 - 8| = 3$,

$$E[X(5)X(8)] = R(3) = 9 + 4e^{-0.2 \times 3} = 9 + 4e^{-0.6}.$$

$$\begin{aligned} \text{Cov}(Z, W) &= E[ZW] - E[Z]E[W] \\ &= (9 + 4e^{-0.6}) - 3 \cdot 3 = 4e^{-0.6} \approx 2.195. \end{aligned}$$

$$\boxed{\mu_x = 3, \quad \text{Var} = 4, \quad \text{Cov}(X(5), X(8)) = 4e^{-0.6}}$$

Problem 4 – Second Moment of $X(8) - X(5)$

Problem 4

For a stationary random process, the ACF is $R(\tau) = Ae^{-|\tau|}$, where A is a constant. Find the second moment of the random variable $X(8) - X(5)$.

Solution

$$R(\tau) = Ae^{-|\tau|} \Rightarrow R(t_1, t_2) = Ae^{-|t_1 - t_2|}, \text{ and } R(0) = Ae^{-0} = A.$$

$$\begin{aligned} E[(X(8) - X(5))^2] &= E[X^2(8)] + E[X^2(5)] - 2E[X(8)X(5)] \\ &= R(0) + R(0) - 2R(8, 5) \\ &= 2A - 2Ae^{-|8-5|} = 2A - 2Ae^{-3} \\ &= 2A(1 - e^{-3}). \end{aligned}$$

$$E[(X(8) - X(5))^2] = 2A(1 - e^{-3})$$

Cross-Correlation Function (CCF) – Property 1**Definition: CCF**

For jointly stationary random processes $x(t)$ and $y(t)$, the CCF is a function of the time difference: $R_{xy}(\tau) = E[x(t)y(t - \tau)]$.

Property 1

$$R_{xy}(-\tau) = R_{yx}(\tau).$$

Proof

$$R_{xy}(-\tau) = E[x(t)y(t + \tau)]. \text{ Let } \lambda = t + \tau \Rightarrow t = \lambda - \tau:$$

$$R_{xy}(-\tau) = E[x(\lambda - \tau)y(\lambda)] = E[y(\lambda)x(\lambda - \tau)] = R_{yx}(\tau).$$

CCF – Property 2**Property 2**

For any two jointly WSS random processes: $|R_{xy}(\tau)| \leq \frac{1}{2}(R_{xx}(0) + R_{yy}(0))$.

Proof

$$\text{We know } R_{xx}(0) = E[x^2(t)], \quad R_{yy}(0) = E[y^2(t)]. \quad (1)$$

Consider $E(x(t_1) + y(t_2))^2 \geq 0$, with $\tau = t_1 - t_2$.

Expanding and using (1):

$$\begin{aligned} E[x^2(t_1)] + E[y^2(t_2)] + 2E[x(t_1)y(t_2)] &\geq 0 \\ R_{xx}(0) + R_{yy}(0) + 2R_{xy}(\tau) &\geq 0 \\ \Rightarrow R_{xy}(\tau) &\geq -\frac{1}{2}(R_{xx}(0) + R_{yy}(0)). \end{aligned} \quad (2)$$

Now consider $E(x(t_1) - y(t_2))^2 \geq 0$:

$$\begin{aligned} E[x^2(t_1)] + E[y^2(t_2)] - 2E[x(t_1)y(t_2)] &\geq 0 \\ R_{xx}(0) + R_{yy}(0) - 2R_{xy}(\tau) &\geq 0 \\ \Rightarrow R_{xy}(\tau) &\leq \frac{1}{2}(R_{xx}(0) + R_{yy}(0)). \end{aligned} \quad (3)$$

Combining (2) and (3):

$$\boxed{|R_{xy}(\tau)| \leq \frac{1}{2}(R_{xx}(0) + R_{yy}(0))}$$

CCF – Property 3

Property 3

$$|R_{xy}(\tau)| \leq \sqrt{R_{xx}(0) R_{yy}(0)}.$$

Proof

Consider

$$E \left(\frac{x(t_1)}{\sqrt{R_{xx}(0)}} + \frac{y(t_2)}{\sqrt{R_{yy}(0)}} \right)^2 \geq 0.$$

Expanding,

$$\begin{aligned} \frac{R_{xx}(0)}{R_{xx}(0)} + \frac{R_{yy}(0)}{R_{yy}(0)} + \frac{2R_{xy}(\tau)}{\sqrt{R_{xx}(0)R_{yy}(0)}} &\geq 0 \\ \Rightarrow 2 + \frac{2R_{xy}(\tau)}{\sqrt{R_{xx}(0)R_{yy}(0)}} &\geq 0 \\ \Rightarrow R_{xy}(\tau) &\geq -\sqrt{R_{xx}(0)R_{yy}(0)}. \end{aligned} \quad (2)$$

Similarly, from

$$\begin{aligned} E \left(\frac{x(t_1)}{\sqrt{R_{xx}(0)}} - \frac{y(t_2)}{\sqrt{R_{yy}(0)}} \right)^2 &\geq 0, \\ 2 - \frac{2R_{xy}(\tau)}{\sqrt{R_{xx}(0)R_{yy}(0)}} &\geq 0 \Rightarrow R_{xy}(\tau) \leq \sqrt{R_{xx}(0)R_{yy}(0)}. \end{aligned} \quad (3)$$

Combining (2) and (3):

$$\begin{aligned} -\sqrt{R_{xx}(0)R_{yy}(0)} &\leq R_{xy}(\tau) \leq \sqrt{R_{xx}(0)R_{yy}(0)} \\ \Rightarrow \boxed{|R_{xy}(\tau)| &\leq \sqrt{R_{xx}(0)R_{yy}(0)}} \end{aligned}$$

CCF – Property 4

Property 4

If $\{x(t)\}$ and $\{y(t)\}$ are two independent (WSS) random processes, then $R_{xy}(\tau) = \mu_x \mu_y$, where $\mu_x = E[x(t)]$, $\mu_y = E[y(t)]$.

Proof

$R_{xy}(\tau) = E[x(t)y(t-\tau)]$. Since the processes are independent, $x(t)$ and $y(t-\tau)$ are independent for all t, τ ; hence

$$R_{xy}(\tau) = E[x(t)] E[y(t-\tau)] = \mu_x \mu_y,$$

because the means of WSS processes do not depend on t .

Problem – CCF of $W(t)$ and $Z(t)$

Problem

Given two stationary random processes $W(t) = A(t) + B(t)$ and $Z(t) = A(t) - B(t)$, where $A(t)$ and $B(t)$ are independent random processes with mean zero and ACFs $R_{AA}(\tau) = e^{-|\tau|}$, $R_{BB}(\tau) = 3e^{-|\tau|}$, find the cross-correlation function of $W(t)$ and $Z(t)$.

Solution

Given $E[A(t)] = 0$, $E[B(t)] = 0$.

$$\begin{aligned} R_{WZ}(\tau) &= E[W(t)Z(t-\tau)] \\ &= E[(A(t) + B(t))(A(t-\tau) - B(t-\tau))]. \end{aligned}$$

$$\begin{aligned} R_{WZ}(\tau) &= E[A(t)A(t-\tau)] - E[A(t)B(t-\tau)] \\ &\quad + E[B(t)A(t-\tau)] - E[B(t)B(t-\tau)]. \end{aligned}$$

By independence and zero means, $E[A(t)B(t-\tau)] = E[A(t)]E[B(t-\tau)] = 0$ and $E[B(t)A(t-\tau)] = 0$. Hence

$$R_{WZ}(\tau) = R_{AA}(\tau) - R_{BB}(\tau) = e^{-|\tau|} - 3e^{-|\tau|} = -2e^{-|\tau|}.$$

$$\boxed{R_{WZ}(\tau) = -2e^{-|\tau|}}$$

Problem – CCF of Two Sinusoids

Problem

Two jointly WSS random processes $X(t)$ and $Y(t)$ have sample functions

$$X(t) = A_1 \cos(\omega_0 t + \Theta), \quad Y(t) = A_2 \cos(\omega_0 t + \Theta + \phi),$$

where Θ is uniformly distributed in $(0, 2\pi)$ and A_1, A_2, ω_0, ϕ are constants. Find the

cross-correlation function $R_{xy}(\tau)$.

Solution

The pdf of Θ is $f_{\Theta}(\theta) = \frac{1}{2\pi}$, $0 \leq \theta \leq 2\pi$.

$$\begin{aligned} R_{xy}(\tau) &= E[X(t)Y(t-\tau)] \\ &= E[A_1 \cos(\omega_0 t + \Theta) A_2 \cos(\omega_0(t-\tau) + \Theta + \phi)] \\ &= A_1 A_2 E[\cos(\omega_0 t + \Theta) \cos(\omega_0 t - \omega_0 \tau + \Theta + \phi)]. \end{aligned}$$

Use $\cos \alpha \cos \beta = \frac{1}{2}[\cos(\alpha + \beta) + \cos(\alpha - \beta)]$ with

$$\alpha = \omega_0 t + \Theta, \quad \beta = \omega_0 t - \omega_0 \tau + \Theta + \phi.$$

$$\begin{aligned} \alpha + \beta &= (\omega_0 t + \Theta) + (\omega_0 t - \omega_0 \tau + \Theta + \phi) = 2\omega_0 t - \omega_0 \tau + 2\Theta + \phi, \\ \alpha - \beta &= (\omega_0 t + \Theta) - (\omega_0 t - \omega_0 \tau + \Theta + \phi) = \omega_0 \tau - \phi. \end{aligned}$$

Let $\psi = 2\omega_0 t - \omega_0 \tau + \phi$, so that $\alpha + \beta = \psi + 2\Theta$. Then

$$\begin{aligned} R_{xy}(\tau) &= \frac{A_1 A_2}{2} E[\cos(\psi + 2\Theta) + \cos(\omega_0 \tau - \phi)] \\ &= \frac{A_1 A_2}{2} (E[\cos(\psi + 2\Theta)] + \cos(\omega_0 \tau - \phi)), \end{aligned}$$

since $\cos(\omega_0 \tau - \phi)$ does not depend on Θ .

$$\begin{aligned} E[\cos(\psi + 2\Theta)] &= \frac{1}{2\pi} \int_0^{2\pi} \cos(\psi + 2\theta) d\theta = \frac{1}{2\pi} \left[\frac{\sin(\psi + 2\theta)}{2} \right]_0^{2\pi} \\ &= \frac{1}{4\pi} (\sin(\psi + 4\pi) - \sin(\psi + 0)) \\ &= \frac{1}{4\pi} (\sin \psi - \sin \psi) = 0. \end{aligned}$$

Therefore

$$R_{xy}(\tau) = \frac{A_1 A_2}{2} (0 + \cos(\omega_0 \tau - \phi)) \Rightarrow R_{xy}(\tau) = \frac{A_1 A_2}{2} \cos(\omega_0 \tau - \phi)$$

Power Spectral Density Function (PSD)

Power Spectral Density Function (PSD)

The PSD describes the distribution of power of the R.P. $\{x(t)\}$ in the frequency domain.

Definition: The PSD is the Fourier transform of the ACF, denoted by $S(\omega)$ or $S_x(\omega)$:

$$S(\omega) = \int_{-\infty}^{\infty} R(\tau) e^{-i\omega\tau} d\tau$$

With $\omega = 2\pi f$, this gives the power distribution over frequency f .

PSD – Inverse Relation

PSD (contd.)

By the inverse Fourier transform:

$$R(\tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S(\omega) e^{i\omega\tau} d\omega$$

Putting $\tau = 0$ ($e^0 = 1$):

$$R(0) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S(\omega) d\omega = E[x^2(t)]$$

Therefore $R(0)$ represents the **average power** of the R.P. $\{x(t)\}$.

Properties of PSD – Property 1

Property 1

$S(\omega)$ is an even function of ω .

Proof

$$S(\omega) = \int_{-\infty}^{\infty} R(\tau) e^{-i\omega\tau} d\tau \Rightarrow S(-\omega) = \int_{-\infty}^{\infty} R(\tau) e^{i\omega\tau} d\tau.$$

Let $\tau = -u \Rightarrow d\tau = -du$; as τ goes from $-\infty$ to ∞ , u goes from ∞ to $-\infty$:

$$S(-\omega) = \int_{\infty}^{-\infty} R(-u) e^{i\omega(-u)} (-du) = \int_{-\infty}^{\infty} R(-u) e^{-i\omega u} du.$$

Since $R(\tau)$ is an even function, $R(-u) = R(u)$:

$$S(-\omega) = \int_{-\infty}^{\infty} R(u) e^{-i\omega u} du = S(\omega).$$

$$\Rightarrow \boxed{S(-\omega) = S(\omega)} \Rightarrow \boxed{S(\omega) \text{ is an even function of } \omega.}$$

Properties of PSD – Property 2

Property 2

$S(\omega)$ is a real function of ω .

Proof

$$\begin{aligned}
S(\omega) &= \int_{-\infty}^{\infty} R(\tau) e^{-i\omega\tau} d\tau = \int_{-\infty}^{\infty} R(\tau) (\cos \omega\tau - i \sin \omega\tau) d\tau \\
&= \int_{-\infty}^{\infty} R(\tau) \cos \omega\tau d\tau - i \int_{-\infty}^{\infty} R(\tau) \sin \omega\tau d\tau.
\end{aligned}$$

Since $R(\tau)$ is even and $\sin \omega\tau$ is odd, $R(\tau) \sin \omega\tau$ is odd, so

$$\int_{-\infty}^{\infty} R(\tau) \sin \omega\tau d\tau = 0.$$

Hence, since $R(\tau) \cos \omega\tau$ is even,

$$S(\omega) = \int_{-\infty}^{\infty} R(\tau) \cos \omega\tau d\tau = 2 \int_0^{\infty} R(\tau) \cos \omega\tau d\tau,$$

which is real.

$$\Rightarrow \boxed{S(\omega) \text{ is a real function of } \omega.}$$

PSD Problem 1 – Triangular ACF**Find the Power Spectral Density (PSD)**

Find the PSD, if the ACF of the random process $\{X(t)\}$ is

$$R(\tau) = \begin{cases} 1 - \frac{|\tau|}{T}, & |\tau| \leq T, \\ 0, & \text{elsewhere.} \end{cases}$$

Solution

By definition, $S(\omega) = \int_{-\infty}^{\infty} R(\tau) e^{-i\omega\tau} d\tau$. Splitting the range:

$$\begin{aligned}
S(\omega) &= \left(\int_{-\infty}^{-T} + \int_{-T}^T + \int_T^{\infty} \right) R(\tau) e^{-i\omega\tau} d\tau \\
&= \int_{-T}^T \left(1 - \frac{|\tau|}{T} \right) e^{-i\omega\tau} d\tau \\
&= \int_{-T}^T \left(1 - \frac{|\tau|}{T} \right) (\cos \omega\tau - i \sin \omega\tau) d\tau.
\end{aligned}$$

The cosine part is even and the sine part is odd, so

$$\begin{aligned}
S(\omega) &= 2 \int_0^T \left(1 - \frac{\tau}{T} \right) \cos \omega\tau d\tau \\
&= 2 \left[\int_0^T \cos \omega\tau d\tau - \frac{1}{T} \int_0^T \tau \cos \omega\tau d\tau \right].
\end{aligned}$$

First integral:

$$\int_0^T \cos \omega \tau d\tau = \left[\frac{\sin \omega \tau}{\omega} \right]_0^T = \frac{\sin \omega T - \sin 0}{\omega} = \frac{\sin \omega T}{\omega}.$$

Second integral, by parts with $u = \tau$, $dv = \cos \omega \tau d\tau \Rightarrow du = d\tau$, $v = \frac{\sin \omega \tau}{\omega}$:

$$\int_0^T \tau \cos \omega \tau d\tau = \left[\frac{\tau \sin \omega \tau}{\omega} \right]_0^T - \frac{1}{\omega} \int_0^T \sin \omega \tau d\tau.$$

$$\begin{aligned} \int_0^T \tau \cos \omega \tau d\tau &= \left(\frac{T \sin \omega T}{\omega} - \frac{0 \cdot \sin 0}{\omega} \right) - \frac{1}{\omega} \left[\frac{-\cos \omega \tau}{\omega} \right]_0^T \\ &= \frac{T \sin \omega T}{\omega} - \frac{1}{\omega^2} (-\cos \omega T + \cos 0) \\ &= \frac{T \sin \omega T}{\omega} - \frac{1 - \cos \omega T}{\omega^2}. \end{aligned}$$

$$\begin{aligned} S(\omega) &= 2 \left[\frac{\sin \omega T}{\omega} - \frac{1}{T} \left(\frac{T \sin \omega T}{\omega} - \frac{1 - \cos \omega T}{\omega^2} \right) \right] \\ &= 2 \left[\frac{\sin \omega T}{\omega} - \frac{\sin \omega T}{\omega} + \frac{1 - \cos \omega T}{T \omega^2} \right] \\ &= \frac{2}{T \omega^2} (1 - \cos \omega T). \end{aligned}$$

$$S(\omega) = \frac{2}{T \omega^2} (1 - \cos \omega T) = \frac{4 \sin^2(\omega T/2)}{T \omega^2}$$

PSD Problem 2 – ACF from PSD

Find the ACF $R(\tau)$

Find the ACF $R(\tau)$ of a stationary R.P. $\{x(t)\}$ whose PSD is

$$S(\omega) = \begin{cases} 1 + \omega^2, & |\omega| \leq 1, \\ 0, & \text{elsewhere.} \end{cases}$$

Solution

$$R(\tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S(\omega) e^{i\omega\tau} d\omega = \frac{1}{2\pi} \int_{-1}^1 (1 + \omega^2) (\cos \omega \tau + i \sin \omega \tau) d\omega.$$

$(1 + \omega^2) \cos \omega \tau$ is even and $(1 + \omega^2) \sin \omega \tau$ is odd, so

$$R(\tau) = \frac{1}{2\pi} \cdot 2 \int_0^1 (1 + \omega^2) \cos \omega \tau d\omega = \frac{1}{\pi} \int_0^1 (1 + \omega^2) \cos \omega \tau d\omega.$$

Integrating by parts (Bernoulli's rule) with

$$u = 1 + \omega^2, \quad u' = 2\omega, \quad u'' = 2,$$

$$v_1 = \frac{\sin \omega \tau}{\tau}, \quad v_2 = -\frac{\cos \omega \tau}{\tau^2}, \quad v_3 = -\frac{\sin \omega \tau}{\tau^3}.$$

$$\begin{aligned} R(\tau) &= \frac{1}{\pi} \left[u v_1 - u' v_2 + u'' v_3 \right]_0^1 \\ &= \frac{1}{\pi} \left[\frac{(1 + \omega^2) \sin \omega \tau}{\tau} + \frac{2\omega \cos \omega \tau}{\tau^2} - \frac{2 \sin \omega \tau}{\tau^3} \right]_0^1. \end{aligned}$$

Substituting the limits:

$$\begin{aligned} \text{at } \omega = 1 : & \quad \frac{2 \sin \tau}{\tau} + \frac{2 \cos \tau}{\tau^2} - \frac{2 \sin \tau}{\tau^3}, \\ \text{at } \omega = 0 : & \quad \frac{(1) \sin 0}{\tau} + \frac{2(0) \cos 0}{\tau^2} - \frac{2 \sin 0}{\tau^3} = 0. \end{aligned}$$

$$\begin{aligned} R(\tau) &= \frac{1}{\pi} \left[\left(\frac{2 \sin \tau}{\tau} + \frac{2 \cos \tau}{\tau^2} - \frac{2 \sin \tau}{\tau^3} \right) - 0 \right] \\ &= \frac{2}{\pi \tau^3} [\tau^2 \sin \tau + \tau \cos \tau - \sin \tau]. \end{aligned}$$

$$R(\tau) = \frac{2}{\pi \tau^3} [\tau \cos \tau + (\tau^2 - 1) \sin \tau], \quad \tau \neq 0$$

At $\tau = 0$:

$$R(0) = \frac{1}{\pi} \int_0^1 (1 + \omega^2) d\omega = \frac{1}{\pi} \left[\omega + \frac{\omega^3}{3} \right]_0^1 = \frac{1}{\pi} \left[\left(1 + \frac{1}{3} \right) - 0 \right] = \frac{4}{3\pi}.$$

PSD Problem 3 – Uncorrelated Samples

Problem 3

Find the ACF $R(\tau)$ when the PSD of a random process $\{x(t)\}$ is

$$S(\omega) = \begin{cases} 1, & |\omega| \leq \omega_0, \\ 0, & \text{elsewhere.} \end{cases}$$

Hence prove that $x(t)$ and $x\left(t + \frac{\pi}{\omega_0}\right)$ are uncorrelated random variables, where $\{x(t)\}$ is a zero-mean WSS process.

Solution

$$\begin{aligned} R(\tau) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} S(\omega) e^{i\omega\tau} d\omega = \frac{1}{2\pi} \int_{-\omega_0}^{\omega_0} e^{i\omega\tau} d\omega = \frac{1}{2\pi} \left[\frac{e^{i\omega\tau}}{i\tau} \right]_{-\omega_0}^{\omega_0} \\ &= \frac{1}{2\pi} \cdot \frac{1}{i\tau} (e^{i\omega_0\tau} - e^{-i\omega_0\tau}). \end{aligned}$$

Using $e^{i\omega_0\tau} - e^{-i\omega_0\tau} = 2i \sin \omega_0\tau$:

$$R(\tau) = \frac{1}{2\pi} \cdot \frac{1}{i\tau} \cdot 2i \sin \omega_0\tau \Rightarrow \boxed{R(\tau) = \frac{1}{\pi} \frac{\sin \omega_0\tau}{\tau}}$$

Step 2: Uncorrelatedness

We must show $\text{Cov}(x(t), x(t + \frac{\pi}{\omega_0})) = 0$, where $\text{Cov}(X, Y) = E(XY) - E(X)E(Y)$.

Since $\{x(t)\}$ has zero mean, $E[x(t)] = 0 = E[x(t + \frac{\pi}{\omega_0})]$, so

$$\begin{aligned} \text{Cov}\left(x(t), x\left(t + \frac{\pi}{\omega_0}\right)\right) &= E\left[x(t) x\left(t + \frac{\pi}{\omega_0}\right)\right] = E\left[x(t) x\left(t - \left(-\frac{\pi}{\omega_0}\right)\right)\right] \\ &= R\left(-\frac{\pi}{\omega_0}\right) = R\left(\frac{\pi}{\omega_0}\right) \quad (R \text{ is even}). \end{aligned}$$

Putting $\tau = \frac{\pi}{\omega_0}$ in $R(\tau) = \frac{1}{\pi} \frac{\sin \omega_0\tau}{\tau}$:

$$R\left(\frac{\pi}{\omega_0}\right) = \frac{1}{\pi} \frac{\sin\left(\omega_0 \cdot \frac{\pi}{\omega_0}\right)}{\pi/\omega_0} = \frac{\omega_0}{\pi^2} \sin \pi = \frac{\omega_0}{\pi^2} \cdot 0 = 0.$$

Therefore

$$\text{Cov}\left(x(t), x\left(t + \frac{\pi}{\omega_0}\right)\right) = R\left(-\frac{\pi}{\omega_0}\right) = R\left(\frac{\pi}{\omega_0}\right) = 0,$$

$$\Rightarrow \boxed{x(t) \text{ and } x\left(t + \frac{\pi}{\omega_0}\right) \text{ are uncorrelated r.v.'s.}}$$

Wiener–Khinchine Theorem

Statement

Let $X_T(\omega)$ be the Fourier transform of the truncated random process

$$X_T(t) = \begin{cases} X(t), & -T \leq t \leq T, \\ 0, & \text{elsewhere,} \end{cases}$$

where $X(t)$ is a real WSS process. Then the PSD is given by

$$\boxed{S(\omega) = \lim_{T \rightarrow \infty} \frac{1}{2T} E[|X_T(\omega)|^2].}$$

Wiener–Khinchine Theorem – Proof

Proof

$X_T(\omega)$ is the Fourier transform of $X_T(t)$:

$$X_T(\omega) = \int_{-\infty}^{\infty} X_T(t) e^{-i\omega t} dt = \int_{-T}^T X(t) e^{-i\omega t} dt. \quad (1)$$

Writing the dummy variable as t_1 and as t_2 :

$$X_T(\omega) = \int_{-T}^T X(t_1) e^{-i\omega t_1} dt_1, \quad (2)$$

$$X_T(\omega) = \int_{-T}^T X(t_2) e^{-i\omega t_2} dt_2. \quad (3)$$

Taking the complex conjugate of (3), and using that $X(t)$ is real ($\overline{X(t_2)} = X(t_2)$):

$$\overline{X_T(\omega)} = \int_{-T}^T X(t_2) e^{i\omega t_2} dt_2. \quad (4)$$

Multiplying (2) and (4):

$$|X_T(\omega)|^2 = X_T(\omega) \overline{X_T(\omega)} = \int_{-T}^T \int_{-T}^T X(t_1) X(t_2) e^{-i\omega(t_1-t_2)} dt_1 dt_2.$$

Taking expectation:

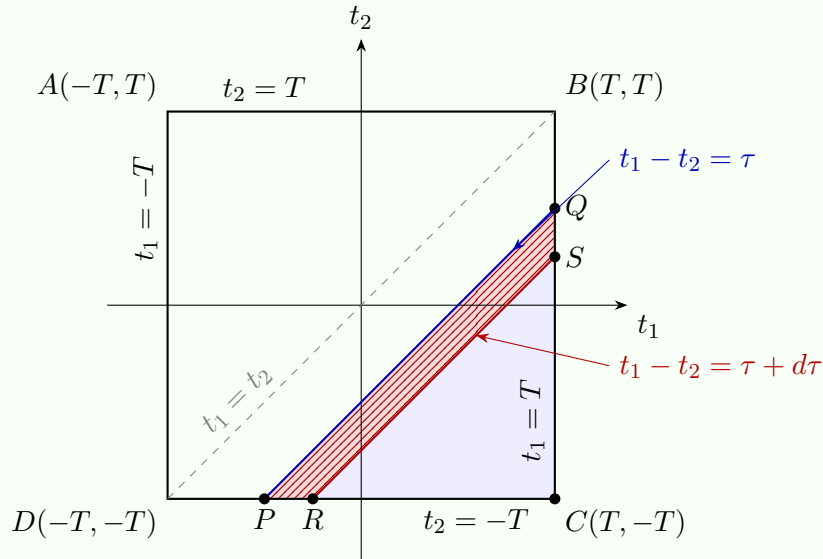
$$E[|X_T(\omega)|^2] = \int_{-T}^T \int_{-T}^T E[X(t_1) X(t_2)] e^{-i\omega(t_1-t_2)} dt_1 dt_2.$$

Since $X(t)$ is WSS, $E[X(t_1) X(t_2)] = R(\tau)$ with $\tau = t_1 - t_2$:

$$E[|X_T(\omega)|^2] = \int_{-T}^T \int_{-T}^T R(\tau) e^{-i\omega\tau} dt_1 dt_2. \quad (5)$$

We now convert this double integral into a single integral in τ .

Region of integration



Here t_1 and t_2 each vary from $-T$ to T , so the region is the square $ABCD$, and $\tau = t_1 - t_2$ varies from $-2T$ to $2T$.

For $0 < \tau < 2T$, the lines $t_1 - t_2 = \tau$ and $t_1 - t_2 = \tau + d\tau$ cut off the thin strip $PRSQ$ (hatched) near the corner C .

Strip $PRSQ = \triangle PQC - \triangle RSC$.

Area of the Elementary Strip

$$\begin{aligned}
dt_1 dt_2 &= \text{area of the elementary strip } PRSQ \\
&= \text{area of } \triangle PQC - \text{area of } \triangle RSC \\
&= \frac{1}{2}(PC)(QC) - \frac{1}{2}(RC)(SC).
\end{aligned} \tag{6}$$

P : intersection of $t_1 - t_2 = \tau$ and $t_2 = -T \Rightarrow t_1 = \tau - T$, i.e. $P(\tau - T, -T)$.

Q : intersection of $t_1 - t_2 = \tau$ and $t_1 = T \Rightarrow t_2 = T - \tau$, i.e. $Q(T, T - \tau)$.

With $C(T, -T)$ (right angle at C):

$$PC = T - (\tau - T) = 2T - \tau, \quad QC = (T - \tau) - (-T) = 2T - \tau.$$

Similarly, R and S lie on $t_1 - t_2 = \tau + d\tau$:

$$\begin{aligned}
R(\tau + d\tau - T, -T), \quad S(T, T - \tau - d\tau), \\
RC = SC = 2T - (\tau + d\tau) = (2T - \tau) - d\tau.
\end{aligned}$$

Hence, from (6),

$$dt_1 dt_2 = \frac{1}{2}(2T - \tau)^2 - \frac{1}{2}[(2T - \tau) - d\tau]^2.$$

Treat $(2T - \tau)$ as a single term and expand $[(2T - \tau) - d\tau]^2 = (2T - \tau)^2 - 2(2T - \tau) d\tau + (d\tau)^2$.

$$\begin{aligned}
dt_1 dt_2 &= \frac{1}{2} \left[(2T - \tau)^2 - \left\{ (2T - \tau)^2 - 2(2T - \tau) d\tau + (d\tau)^2 \right\} \right] \\
&= \frac{1}{2} \left[(2T - \tau)^2 - (2T - \tau)^2 + 2(2T - \tau) d\tau - (d\tau)^2 \right] \\
&= \frac{1}{2} \left[2(2T - \tau) d\tau - (d\tau)^2 \right] = (2T - \tau) d\tau - \frac{1}{2}(d\tau)^2.
\end{aligned}$$

Since $d\tau$ is infinitesimal, $(d\tau)^2$ is of higher order and is neglected:

$$dt_1 dt_2 = (2T - \tau) d\tau, \quad 0 \leq \tau \leq 2T.$$

The Case $\tau < 0$

For $-2T \leq \tau < 0$, the strip between $t_1 - t_2 = \tau$ and $t_1 - t_2 = \tau + d\tau$ lies near the corner $A(-T, T)$. Its intercepts on $t_1 = -T$ and $t_2 = T$ give legs

$$2T + \tau = 2T - |\tau|,$$

and exactly the same computation gives $dt_1 dt_2 = (2T + \tau) d\tau$.

Combining both cases:

$$\boxed{dt_1 dt_2 = (2T - |\tau|) d\tau, \quad -2T \leq \tau \leq 2T.}$$

Substituting into (5):

$$E[|X_T(\omega)|^2] = \int_{-2T}^{2T} R(\tau) e^{-i\omega\tau} (2T - |\tau|) d\tau.$$

Dividing both sides by $2T$:

$$\begin{aligned}
\frac{1}{2T} E[|X_T(\omega)|^2] &= \int_{-2T}^{2T} R(\tau) e^{-i\omega\tau} \left(\frac{2T - |\tau|}{2T} \right) d\tau \\
&= \int_{-2T}^{2T} R(\tau) e^{-i\omega\tau} \left(1 - \frac{|\tau|}{2T} \right) d\tau.
\end{aligned}$$

Taking the Limit $T \rightarrow \infty$

For each fixed τ ,

$$\lim_{T \rightarrow \infty} \left(1 - \frac{|\tau|}{2T}\right) = 1 - \frac{|\tau|}{\infty} = 1 - 0 = 1,$$

and the limits of integration become $\lim_{T \rightarrow \infty} (-2T) = -\infty$, $\lim_{T \rightarrow \infty} (2T) = \infty$. Hence (for $R(\tau)$ absolutely integrable)

$$\lim_{T \rightarrow \infty} \frac{1}{2T} E[|X_T(\omega)|^2] = \int_{-\infty}^{\infty} R(\tau) e^{-i\omega\tau} \cdot 1 \, d\tau = \int_{-\infty}^{\infty} R(\tau) e^{-i\omega\tau} \, d\tau.$$

Conclusion

By the definition of the power spectral density,

$$S(\omega) = \int_{-\infty}^{\infty} R(\tau) e^{-i\omega\tau} \, d\tau.$$

Therefore

$$S(\omega) = \lim_{T \rightarrow \infty} \frac{1}{2T} E[|X_T(\omega)|^2]$$

which completes the proof of the Wiener–Khinchine theorem.